

Nir Pechuk

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EDUCATION

Carnegie Mellon University

Pittsburgh, PA

B.S. in Computer Science | GPA: 4.0/4.0 | Dean's List, High Honors

Expected Graduation: 05/29

- **Completed Coursework:** Inference Algorithms for Language Modeling (11-763, Sit-in), Linear Algebra (21-241), Vector Calculus (21-266), Computer Systems (15-213), Imperative Computation (15-122), Functional Programming (15-150), Discrete Math (15-151).
- **In-Progress Coursework:** AI/ML I (07-280), Computer Vision (16-385), Probability & Computing (15-259), Parallel & Sequential Data Structures & Algorithms (15-210), Type Theory (98-317).
- **Research Chair, ACM@CMU:** directing research projects and seminar for CMU's largest CS organization.

RESEARCH

Research Assistant – Generative Simulation for Robot Learning

Jan 2026 – Present

Prof. Zackory Erickson's RCHI Lab, Carnegie Mellon University

- Developing GenSimPHRI, a “text2sim2real” framework that turns short text task descriptions into zero-shot sim-to-real policies for physical human-robot interaction tasks, trained on large-scale synthetic datasets.
- Proposed and implemented a residual imitation learning method that improved completion and efficiency on long-horizon human-robot interaction tasks by 50%.

Research Lead – Assistive Sensing Systems

2020 – 2025

Extentek

- Invented and developed a LiDAR ToF-based device to prevent liquid overpouring for visually impaired users.
- Granted US Patent 10830626 B2; published research in [IOP Science](#) on material level indication .

EXPERIENCE

Teaching Assistant

Jan 2026 – Present

Carnegie Mellon University School of Computer Science

Pittsburgh, PA

- Computer Systems (15-213, Fall 2026): low-level systems/performance programming & debugging in C.
- Imperative Computation (15-122, Spring 2026): data structures & algorithms in C.

Software Engineer Intern

May 2026 – August 2026

Palantir Technologies

Seattle, WA

- Re-architected a core production service to reduce memory footprint by 10x and enable horizontal scaling.
- Then, led a team of three engineers on a major refactor of a fleetwide service.

Software Engineer Intern

Summer 2025

Firebolt Inc.

Seattle, WA

- Trained an RNN to predict AWS EC2 capacity and customer demand in tandem, informing compute acquisition.
- Deployed the ML pipeline to a Go gRPC microservice, reducing compute costs by 25% and preventing failure.

SELECTED PROJECTS

Phagentic

- Built a physical bioreactor from scratch (ESP32-driven stirrer, dosing pumps, RGB sensor).
- Trained an MPC-based control policy that reads live sensor state and actuates the hardware in real time to sustain target oscillation amplitude and period of the chemical reaction.

Ballot Guide | Winner out of 270+ projects, CMU TartanHacks 2026

- Built a custom RAG pipeline integrated with a self-scraped vector database of 30,000 legislation entries.
- Developed custom 3D visualizations to demonstrate the local impact of election policies to users.

Truely | Winner out of 350+ projects, UC Berkeley AI Hackathon 2025

- Built a multi-agent LLM pipeline that scrubs the live DOM via a custom browser extension.
- Coordinated web-search verification models to flag misinformation and deepfakes with automatic inline citations.

TECHNICAL SKILLS

Languages: C, Go, C++, Python, Java, Swift, JavaScript/TypeScript, SQL, HTML/CSS

ML/Robotics Frameworks: PyTorch, ROS2, NumPy, Pandas, Gemini/OpenAI/Anthropic API